

NATIONAL OPEN DATA INNOVATION CHALLENGE 2 — ENERGY & ENVIRONMENT

GridIQ — Solutions Report

Team Eukaryote1.0 | Gaborone, Botswana | September 2026

Plan the grid as one system, not three separate bets.

Anchor: 642 GWh of electricity was lost in FY2023 — 14.51% of units — by BPC’s own accounting.

Executive Summary

Botswana plans its electricity system as a series of separate projects. Generation, storage and grid are decided one at a time, so the country stays over-reliant on coal at Morupule and on imported power, rural villages wait for a last mile that is never the cheapest option, and the network is run without knowing which assets matter most.

GridIQ is one shared, data-grounded model of that system, with four engines reading the same topology, demand and injection points:

1. **System-wide optimisation** — what to build, where, how much and how to connect, against one objective.
2. **Flow, congestion & losses** — where constraints bind and what they cost.
3. **Asset-risk screening** — which assets are structurally critical, plus the data contract that would unlock prediction.
4. **VillageFit** — the right technology for each unconnected village, scored against the cost of a line.

Because all four read the same substrate, no recommendation can silently contradict another. This is what makes GridIQ a platform rather than four scripts.

Our differentiator is that the model’s loss output is **checkable against a figure BPC has already published** (642 GWh, FY2023). So the demo is a falsifiable test, not a presentation: if the numbers reconcile, every other engine inherits the credibility of a substrate that just proved itself.

One-sentence pitch: GridIQ helps BPC and Ministry planners — and the households and villages they serve — who struggle with a grid planned project-by-project, by providing one shared model of generation, storage, grid and demand, so that supply is planned at the lowest system cost, losses fall, and the last mile is served by the right technology.

1. Problem Definition & Context

1.1 Supply is planned project-by-project

Botswana’s fleet is dominated by two coal stations — Morupule A (132 MW) and Morupule B (600 MW) — plus diesel peaking. Installed capacity sits around 892 MW against peak demand of 678 MW (2024), yet on a five-year rolling average domestic generation has covered only about 62% of supply. The rest is imported, and in Q1 2026 80.9% of imports came from Eskom alone. Local generation has covered as little as ~44% of distribution in a quarter. Meanwhile, new projects are appraised on resource potential alone: a strong solar site far from load can force a costly line upgrade the resource map never shows.

Who is affected	How they are harmed
Households and businesses BPC and Ministry planners	Unreliable supply and import-priced electricity. Must choose what to build next with no system-cost view.

Who is affected	How they are harmed
Budget and tariff payers	Grid spend that a better-placed project would not have needed.

Failure mode	Consequence	Scale
One dominant generation site	A single coal fleet's availability sets national supply.	National
One dominant import corridor	Cost and reliability exposed to one supplier.	National
Project-by-project appraisal	Stranded grid spend; sub-optimal siting.	Every new project

1.2 The network is operated reactively

There is limited published visibility of where the network is congested, or which assets have no redundant path. Faults are answered after they happen. Public asset age, condition, inspection and failure data for BPC and other Botswana utilities is limited: no open dataset was found.

Who is affected	How they are harmed
Households and businesses	Outages they cannot anticipate or avoid.
BPC crews	Respond to failures instead of pre-empting them.
Planners	Cannot rank where maintenance or redundancy pays.

Failure mode	Consequence	Scale
No criticality view	A non-redundant asset can island a load pocket.	Network-wide
No condition data	Maintenance is reactive, not prioritised.	Every asset

1.3 The last mile is stuck

As of October 2024, 463 of Botswana's 565 gazetted villages were connected — 102 still unconnected, roughly one in six. These are exactly the low-density, far-flung settlements skipped in earlier phases, where line extension is disproportionately expensive. Government's stated fix is solar off-grid electrification, applied as the default rather than chosen per village.

Who is affected	How they are harmed
Rural households	No grid power; limited study, enterprise and health services.
Rural students	A government laptop with nowhere to charge it.
The state budget	Pays for a generic rollout instead of the cheapest fit.

Failure mode	Consequence	Scale
One technology prescribed everywhere	Poor fit where sun is weak and wind/biomass is strong.	~100 villages

Failure mode	Consequence	Scale
No per-village comparison	Line extension assumed, not tested against mini-grids.	~100 villages

2. Solution Description

2.1 The shared substrate

GridIQ builds one network model of Botswana: buses (substations, generation, import gateways, load nodes) and edges (400/220/132 kV lines), with district demand, access gaps and resource potential attached. Every engine reads this same substrate, so a siting recommendation from Engine 1 can be re-priced by Engine 2 and re-located by Engine 4 without contradiction. *(The topology is drawn from OpenStreetMap/OpenInfraMap — the closest open substitute for BPC’s own grid data — cross-checked against the World Bank/AICD Botswana grid layer (2006-vintage, which captures only ~40% of the current 400/220/132 kV network), and patched with BPC’s public post-2021 additions: North West Phase 1, Mochudi and Tlokweng.)*

2.2 Engine 1 — System-wide optimisation

- **What it is:** a screening capacity-expansion model over representative periods.
- **What it computes:** the combination of generation, storage and grid capacity that delivers demand at the lowest total system cost — minimising imports plus capital cost plus losses.
- **Inputs:** topology, district demand, solar/wind resource, existing fleet, import volumes, candidate sites, regional cost benchmarks.
- **User-facing artifact:** a ranked build plan — site, technology, MW, connection point and projected import reduction per Pula.

2.3 Engine 2 — Flow, congestion & losses

- **What it is:** a linearised DC power-flow / OPF model over the real topology.
- **What it computes:** where flow concentrates, which constraints bind, loss per line, and the redispatch cost when a constraint forces costly supply. *(Power is not “routed”: current divides across all parallel paths by impedance, so we model constraints, not shortest paths.)*
- **Inputs:** topology, assumed r/x by voltage class, demand, injections.
- **User-facing artifact:** a loss and congestion panel, reconciled against BPC’s published figure (642 GWh, FY2023; World Bank 15.35%, 2021; AFREC ~630 GWh / 13.8%, 2023). Scope is stated openly: our model is transmission-level, BPC’s figure is T&D.

2.4 Engine 3 — Asset-risk screening, with a data contract

- **What it is:** topology-based criticality, not a prediction model.
- **What it computes:** a ranked watch-list of lines, substations and transformers by structural criticality (no redundant path) and estimated stress.
- **Inputs:** topology alone.
- **User-facing artifact:** a ranked watch-list plus a one-page data contract listing the exact BPC fields (age, condition, inspections, failure history, ratings, impedances) that would upgrade the proxy into prediction.

2.5 Engine 4 — VillageFit

- **What it is:** a per-village technology recommender.
- **What it computes:** the best-fit mix of solar, wind, biomass, biogas or hybrid for each unconnected village, scored against the cost of extending the line — including how the village uses power (e.g. evening study, where solar alone fails).

- **Inputs:** settlement layer (OSM/HDX), population density, Solar Atlas/NASA POWER, livestock and crop data, distance to grid, terrain.
- **User-facing artifact:** an off-grid shortlist with right-sized systems, and a “no-build” alternative fed back into Engine 1.

Integration note: VillageFit is activated from the same substrate as the national engines. A village’s power is not a separate rural project; it is a system-cost decision the national plan currently makes project-by-project — and GridIQ makes it comparable.

2.6 The decision GridIQ is designed to change

These are reasoned inferences from the data and policy context, not measured findings.

Today, BPC’s transmission-planning engineer must decide whether a proposed solar plant can connect to an existing corridor, using resource maps and manual experience, but there is **no public view** of network headroom or of the system-cost consequence of one connection option versus another. (*Grounding: the OIC2 hub names the grid planner as this track’s target user, and BPC grid and load data are not openly available.*)

GridIQ combines the transmission topology, district demand and resource layers to **estimate** the binding constraint and the redispatch/loss cost of each option, under stated line-rating assumptions.

This allows the decision to be made on system cost, not resource potential.

Today, a student in an unconnected village like Gomolemo must decide how to study in the evening, and there is **no published comparison** of whether his village is better served by a line or a mini-grid. **GridIQ’s VillageFit** scores each unconnected village’s own resource mix against line-extension cost. (*Representative scenario based on documented evidence — 102 unconnected villages, 14.3% access in Ngwaketse West — not a documented individual; the laptop detail is illustrative.*)

3. Data & Technical Explanation

3.1 Data sources

Taxonomy: **REAL** = confirmed in the OIC2 hub; **EXTERNAL** = open Botswana/utility source outside the hub, cited directly; **BENCHMARK** = regional or methodological analogue, not Botswana data; **DEMANDED** = not openly available as of Sept 2026 (we call for its release); **ASSUMED** = model assumption, labelled.

Dataset	Source	Type
Electricity generation & distribution by source, quarterly 2016–2026	Statistics Botswana	REAL
National Energy Use Survey 2022/23 (district demand, access)	Statistics Botswana	REAL
Africa Energy Balance 2025 (plant efficiency, losses)	AFREC, African Union	REAL
OpenInfraMap & Global Power Plant Database	OpenInfraMap / WRI	REAL
Global Solar Atlas, Global Wind Atlas, NASA POWER	World Bank/DTU; NASA	REAL
SAPP load & interconnection-capacity reports	SAPP	REAL

Dataset	Source	Type
2022 Census, sub-district shapefile, Meta population density, OSM/HDX extracts	Statistics Botswana / Meta / OSM	REAL
Agricultural Census 2025 (livestock), FAO livestock production	Statistics Botswana / FAO	REAL
BPC Integrated Annual Report 2023 (T&D losses)	BPC	EXTERNAL
Botswana Electricity Transmission Network (2006, cross-check)	World Bank / energydata.info	EXTERNAL
Botswana techno-economic dataset (T&D and plant costs)	Zenodo / Loughborough CCG	EXTERNAL
Eskom Data Portal (import-side benchmark)	Eskom	EXTERNAL
Africa Electricity Transmission & Distribution Grid Map (2017)	World Bank / ESMAP	BENCHMARK
gridfinder predicted global grid	World Bank / ESMAP	BENCHMARK
SciGRID / GridKit / POMATO line parameters (r, x by voltage)	Open research	BENCHMARK
Global Electrification Platform / OnSSET	World Bank / ESMAP	BENCHMARK
ENTSO-E Transparency Platform / AEMO	Europe / Australia	BENCHMARK
Grid infrastructure & demand data (ratings, loads)	BPC	DEMANDED
Grid generation statistics & average demand	BOCRA	DEMANDED
Business register: large industrial consumers	Statistics Botswana	DEMANDED
Line ratings/impedances	Benchmarked from SciGRID/POMATO defaults	ASSUMED

3.2 Regional & benchmark sources

Where Botswana does not publish a layer, we use labelled analogues rather than leaving a blank — never presenting them as Botswana data. The **Africa grid map** (World Bank/ESMAP) cross-checks our OSM topology and adds cross-border corridors. **SciGRID** / **GridKit** / **POMATO** provide default per-km resistance and reactance by voltage class, so assumed line parameters are calibrated rather than guessed. The **Global Electrification Platform** / **OnSSET** supplies the least-cost electrification method for Engine 4. **ENTSO-E** and **AEMO** are methodological analogues for load shapes, congestion and loss factors. For context: other countries do publish full open grid views — the EU mandates it (ENTSO-E), and the US and Australia publish transmission and constraint data — so Botswana’s gap is one of publication, not possibility.

3.3 Technology stack

Layer	Technology	Role
Geospatial database	PostgreSQL + PostGIS + pgRouting	Substrate of record: topology, demand, resources, scenario results.
API	FastAPI + GeoAlchemy2	Graph queries, scenario management, validation endpoints.
Job queue	Celery + Redis	Runs optimisation, power-flow and siting jobs off-request.
Engine 1	linopy + HiGHS	LP capacity-expansion screening.
Engine 2	PyPSA / pandapower	DC power flow, congestion, losses.
Engine 3	networkx / igraph	Criticality, bridges, N-1 islanding.
Engine 4	GeoPandas + rasterio	Raster sampling and per-village scoring.
Frontend	Next.js + MapLibre / Deck.GL	Map layers, panels, what-if slider.
Tiles	pg_tileserv + TiTiler	Vector tiles from PostGIS; COG rasters for the Solar Atlas.
Data integration	Overpass API, NASA POWER API, SAPP, Stats Botswana portal	Reproducible ingestion with provenance.

3.4 How the system works end-to-end

1. **Ingest** — open datasets are pulled and normalised; every row carries a provenance tag (measured, inferred, assumed).
2. **Substrate** — topology is built, cleaned (endpoints snapped, duplicates merged, connectivity checked) and patched with post-2021 additions.
3. **Layers** — demand is allocated to buses via population density; resources are attached per node and district.
4. **Engines** — E2 runs first and its loss output is reconciled against 642 GWh. Only then do E1 (build plan), E3 (criticality) and E4 (VillageFit) run on the validated substrate.
5. **Scenarios** — every dashboard slider is a parameter bundle, so “what-if: add 100 MW here” never requires re-coding.
6. **Presentation** — the map, the loss-reconciliation panel and the build plan are served to the dashboard.

3.5 What is real vs simulated

Category	Content
Real (open data)	Topology, generation fleet, district demand, access rates, resource layers, historical generation/import/loss series.
External (open, outside hub)	BPC loss series; AICD/BPA 2006 grid layer; Botswana techno-economic costs and T&D parameters.
Benchmarked (analogue)	Africa grid map, gridfinder, SciGRID/POMATO line parameters, GEP/OnSSET method, ENTSO-E/AEMO methods.

Category	Content
Assumed (labelled)	Line ratings/impedances, bus-level demand allocation, future demand growth.
Simulated (demo)	Scenario outputs — build plans, criticality rankings and off-grid shortlists are model products, not measured results.

4. Implementation & Next Steps

4.1 Current status

- Problem validation completed against named open datasets (this brief).
- Architecture fixed: one substrate, four engines, Option B stack.
- Data mapped and verified in the OIC2 hub, including one external anchor (BPC 642 GWh).
- Regional/benchmark data ingested and validated: the World Bank/AICD Botswana grid layer (cross-check) and a Botswana techno-economic dataset (costs and T&D parameters, Zenodo). Provenance recorded in `datasets/PROVENANCE.md`.
- Prototype: substrate build and the loss-reconciliation panel are the first build targets at the hackathon.

4.2 Limitations and mitigation

Limitation	Impact	Mitigation
Line ratings/impedances are not openly published	Constraints are estimated.	Calibrate from SciGRID/POMATO defaults; report a bracketed range; publish a data contract for BPC.
Substation-level demand is limited (not openly published)	Demand is allocated, not measured.	Dasymetric allocation from NEUS + population density, labelled an assumption.
Failure history is not openly available	Prediction is not supportable today.	Criticality screening only; state it plainly.
Botswana-specific capex data is limited	Build-plan costs are indicative.	Use the Botswana techno-economic dataset (T&D and plant costs) plus regional benchmarks, all labelled.
OSM topology predates new lines	Missing post-2021 assets.	Manual patch (North West Phase 1, Mochudi, Tlokweng); AICD/BPA layer as cross-check.
Representative periods, not 8,760 h	Not a full expansion model.	Position as a screening tool; do not imply more.
Model is transmission-level; BPC losses are T&D	Direct reconciliation is approximate.	Report the transmission bracket against BPC's T&D total and explain the scope.
Topology is OSM/AICD-derived, not BPC's own	Approximate; not enough for precise power flow without ratings.	State it openly; the data contract is the path to BPC's real model.

Limitations we accept. We would rather name these than have a judge find them. The topology is an open approximation, not BPC's own; line parameters are benchmarked; demand is allocated from

district data; there is no hourly Botswana load series, so representative periods use labelled analogue profiles; and the distribution network is modelled only as an aggregate. None of these stop the falsifiable test we lead with — reconciling modelled losses against BPC’s published 642 GWh — and each has a named path to improvement.

4.3 Roadmap

Phase	Timeline	Activities	Success metric
1. Substrate + test	Hackathon Day 1	Ingest, build/clean topology, allocate demand, run loss reconciliation.	Modelled loss within a stated bracket of 642 GWh.
2. Engines + interface	Hackathon Days 2–3	E2 flow/congestion, E3 criticality, E4 siting, E1 screening; map and panels.	All four engines run on one substrate; dashboard live.
3. Pilot (proposed)	12 weeks post-hackathon	Reconcile with BPC; one-region screening; agree the data contract.	Deployment-ready pilot plan with partner metrics.

4.4 Partners and required resources

- **Botswana Power Corporation** — grid data, loss reconciliation, data contract.
- **Ministry of Minerals & Energy** — planning context and policy alignment.
- **BERA** — regulatory data and tariff context.
- **Statistics Botswana** — demand, census and settlement data.
- **SAPP** — interconnection capacity and regional trade options.
- **BDIH / organisers** — mentorship, government introductions, post-hackathon support.
- **In-country renewable developers** — delivery partners for whatever the siting engine recommends.
- **BIUST** — pending climate-risk and land-cover layers for siting.

5. Impact & The Ask

One anchor number. 642 GWh of electricity was lost in FY2023 — **14.51% of units** — by BPC’s own Integrated Annual Report 2023, cross-checked by the World Bank (15.35%, 2021) and AFREC (~630 GWh, 13.8%, 2023). GridIQ’s first job is to reproduce that number from open data, because a model that can reproduce the losses BPC already reports can be trusted on the decisions it recommends.

One scale sentence. GridIQ scales by planning the existing grid as one system, with every recommendation traced to a named open dataset — no new data collection required to start.

One 12-week ask. Over 12 weeks, working with BPC and Statistics Botswana, we will reconcile the model’s loss output against BPC’s published 642 GWh, complete a system-cost screening for one region’s next connection decision, and agree the data contract that turns asset-risk screening into prediction — producing a deployment-ready pilot plan with partner-ready metrics.

6. AI Disclosure

AI tools were used to assist with research synthesis, document drafting and code scaffolding. Every data claim in this report is cited to a named source, and the team has reviewed and owns every figure and design decision. No confidential or personal data was used.